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Palaeomagnetism of Lower Devonian units from Gaspé, Quebec

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Palaeomagnetic studies are reported from fifty oriented samples (117 specimens) at 10 sites in the eastern Gaspé Peninsula. All sampled lithostratigraphic units are composed of siltstones and sandstones of Devonian age. The Indian Point Formation is of Gedinnian age and the York River and Battery Point Formations of Emsian age. At most of the sites sampled the sedimentary formations are cut by Middle to Late Carboniferous (mean K-Ar age = 300 Ma) diabase dykes which were also sampled. Three components of magnetization were isolated. The most common component A ($D = 313°$, $I = -45°$) is present in 8 sites and in all Indian Point and Battery Point sites. Component B ($D = 147°$, $I = 38°$) is found in 3 sites. Component B is antiparallel to component A and their mean direction is $D = 317°$, $I = -43°$; this value is similar to the characteristic remanence ($D = 329°$, $I = -43°$) of the dykes. Component C is found mainly in the York River Formation (3 sites, $D = 182°$, $I = 41°$). Contrary to component A-B, component C passes (barely) the fold test with a tilt-corrected direction of $D = 174°$, $I = 75°$; it also passes baked contact tests. Since component C is pre-folding, it is older than Late Devonian (Acadian Orogeny) but may be primary or secondary. Its palaeopole is 109°E, 20°S and it is characterized by a high palaeolatitude (62°S). Since the palaeopole of component A-B is 154°E, 8°N, the APW trajectory moved in the northeast direction during the Lower/Middle Devonian to the Middle/Upper Carboniferous time interval.

1. Introduction

In spite of extensive palaeomagnetic studies over the last 30 years, the Lower Palaeozoic apparent polar wander path for North America is poorly defined. A great many pole positions from Silurian and Devonian rock units lie within 20° of the Permo-Carboniferous poles. This implies that either North America moved very little relative to the palaeopoles in this time interval or else that Permo-Carboniferous overprints predominate [1]. Since a large number of the palaeomagnetic studies originate from the Appalachians, enough data has become available to compare them with those obtained on cratonic North America. Apparent latitudinal discrepancies between the two sets of data led to the hypothesis of sinistral strike slip motion [2] and/or rotation [3] of terranes within the Appalachians. Recent studies [4–6] have demonstrated that this first hypothesis is very unlikely but the second may be applicable in some instances.

For the Devonian period, a great many palaeomagnetic data are available in North America. However, the large scatter of palaeopoles and palaeolatitudes is such that an interpretation of all these palaeomagnetic results is a difficult task. Many attempts were made to obtain reliable Devonian palaeopole positions and apparent polar wander path (APWP) for cratonic North America (e.g. [7–9]). For various reasons (e.g. insufficient data coverage, magnetic overprints, inaccurate dating, etc.), the Devonian APWP for the North American craton is not well defined. For instance, Van der Voo [8] concluded at a small northward oriented APWP along longitude 105°E for this period while Seguin [9] arrived at a loop centered at latitude 10°N and bounded by latitudes 90° and 145°E. Obviously, the first interpretation implies little or very little polar motion and the second a large one. Establishing a definite Devonian APWP reference frame for the North American craton is of prime importance in order to quantify the direction and rate of displacements of the terranes (e.g. Andean margin of Iapetus, Avalon, Meguma) accreted to the eastern margin of North America. In order to do this, a systematization is needed and Devonian palaeo-

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poles must be subdivided in their respective regions (terranes) as attempted by Seguin [9] and Seguin and Gahé [10].

The same authors concluded that by Early to Middle Devonian time, the Andean margin of Iapetus (Humber, Dunnage and Gander zones of the northern Appalachians) defined an APW loop similar to the one obtained for cratonic North America but slightly displaced towards the eastern latitudes. In other words, the palaeolatitudes of the Andean margin of Iapetus are located 10–15° to the south of those obtained for cratonic North America. According to Seguin [9], the APW loop for the Avalon Zone is the same as for the terranes in the Andean margin of Iapetus in Middle and Late Devonian time but different for the Early Devonian interval.

In order to define accurately the APWP of North America, well-dated components of magnetization must be retrieved. Only positive field tests such as fold, conglomerate or baked contact tests permit definition of the relative age of acquisition of a magnetization. The paleomagnetic study reported here comes from the Lower Devonian sediments of the Gaspé Peninsula and was carried out with these objectives in mind.

2. Geological setting

The region studied is located in the Gaspé–Connecticut Valley synclinorium of the Quebec Appalachians. The geological information in this area was obtained in three main phases: (1) regional mapping [11,12], (2) systematic mapping [13,14], and (3) detailed studies [15–22] which included stratigraphic, sedimentological and palaeontological investigations. The Indian Point Formation represents the upper part of the Chaleurs Group and spans the Upper Silurian/Lower Devonian (Emsian) interval. This group overlies the Matapedia Group (mainly Ordovician) and the Quebec Group (Cambrian and Lower Ordovician). The Chaleurs Group is overlain by Lower Devonian Gaspé limestones (Siegenian age) and Gaspé sandstones (Emsian age); the York River and Battery Point Formations constitute the central units of this last group which is sometimes unconformably overlain by Carboniferous red beds (sandstones and conglomerate) and volcanics and is cut by Carboniferous diabase dykes. Except for the Carboniferous, all lithological units were deformed (mainly folded, sometimes faulted) during the Acadian Orogeny in Late Devonian time.

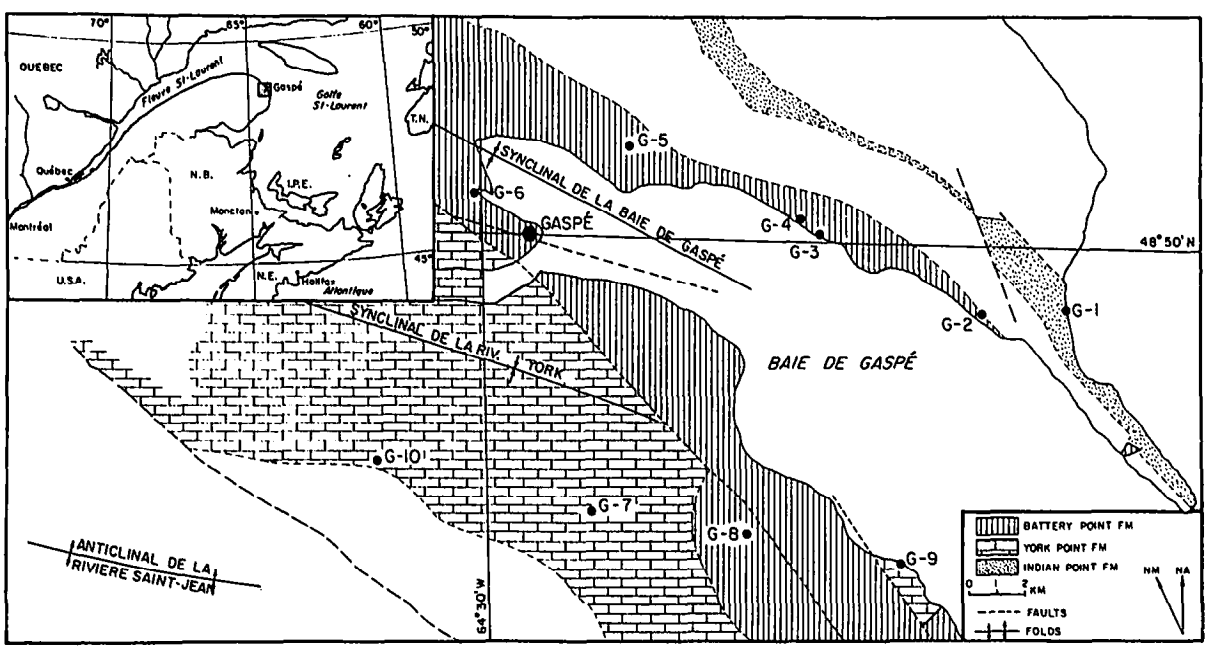


Fig. 1. Location map and simplified geological map of the eastern Gaspé area, province of Quebec. The sampling localities are indicated and numbered. Map modified after Brisebois [20,21].

In the region studied, the Indian Point Formation is subdivided into five distinct members. The member sampled is mainly composed of somewhat limey siltstone and fine-grained sandstone; a few beds of limey sandstones and limestones were also observed. The thick-bedded greenish volcaniclastic sandstone was sampled in this formation (Fig. 1). The York River Formation is composed of conglomerate sandstone, siltstone and mudstone. The sampling was done preferentially in the sandstone and occasionally in the siltstone. The sandstone is greyish green, fine- to medium-grained and is characterized by a limey matrix in some instances. The Battery Point Formation includes three members: Petit-Gaspé, Cap-aux-Os and Fort Prével. The two first members were sampled; sites G2, G6 and G8 were obtained from the first member and sites G3, G4 and G5 from the second. The Petit-Gaspé member is composed of thick-bedded medium- to coarse-grained sandstone with occasional conglomeratic layers and thin dolomitic beds in a few places. The medium-grained sandstone of this member was sampled. The Cap-aux-Os member is distinguished from the previous member by the occurrence of red siltstones and mudstones as well as sequences of sandstones with smaller grain size. The strike of the formations is E-W and the dips vary from 65° N to 40° S.

The diabase dykes cut vertically the Lower Devonian formations; their thicknesses vary between 5 and 10 m. The diabase is composed of plagioclase and pyroxene with an ophitic texture; they are often vacuolate [20]. The age of these diabases was determined isotopically by whole-rock K-Ar method [23,24]. Samples GSC 65-126, 65-127 and 72-99 were taken in the diabases and yielded respective ages of 319, 295 and 294 Ma. Samples GSC 72-101, and 72-102 were taken in basalts associated with the dykes and gave respectively 292 and 291 Ma. A more recent whole-rock K-Ar age was obtained on a trachyandesite in the Sunny Bank drill hole (R. Laurent, personal communication); its age is 308 Ma. The mean age of the diabase dykes is 300 Ma, i.e. Upper Carboniferous.

3. Palaeomagnetic sampling and measurements

In the present palaeomagnetic study, 50 oriented samples (117 specimens) at 10 sites were collected from well studied and dated localities. The sampling was carried out with a portable diamond drill and orientations were measured with a Brunton compass. Because of good exposures, sampling could be made at baked margins of country rocks associated with contacts of diabase dykes in order to perform palaeomagnetic baked contact tests.

Magnetization was measured with a digital Schonstedt spinner magnetometer (DSM-1). Two methods of demagnetization were carried out using: (1) a Schonstedt thermal demagnetizer (TSD-1) in residual fields of less than 15 nT, and (2) a GSD-1 Schonstedt alternating field demagnetizer (mean intensity = 100 mT). The estimated uncertainties in the NRM determinations are about 6% in intensity and 4° in direction.

4. NRM directions and intensities

Fig. 2 shows the directions of natural remanent magnetization (NRM) for all samples collected on the Indian Point, York River and Battery Point Formations. The NRM directions are mainly oriented in a NW-SE subvertical plane and about 15% of them coincide with the present Earth's

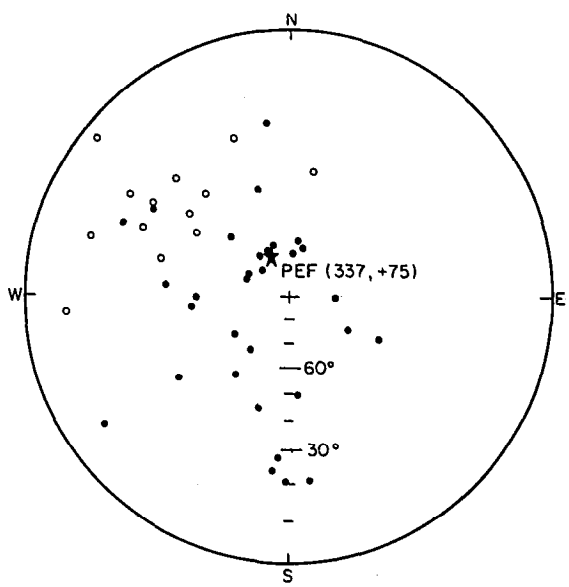


Fig. 2. Sample NRM directions plotted on Wulff net for the sedimentary rocks of the three geological formations sampled. Solid (open) circles represent projections on the lower (upper) hemisphere. The star represents the present Earth's field (PEF) of the sampling region.

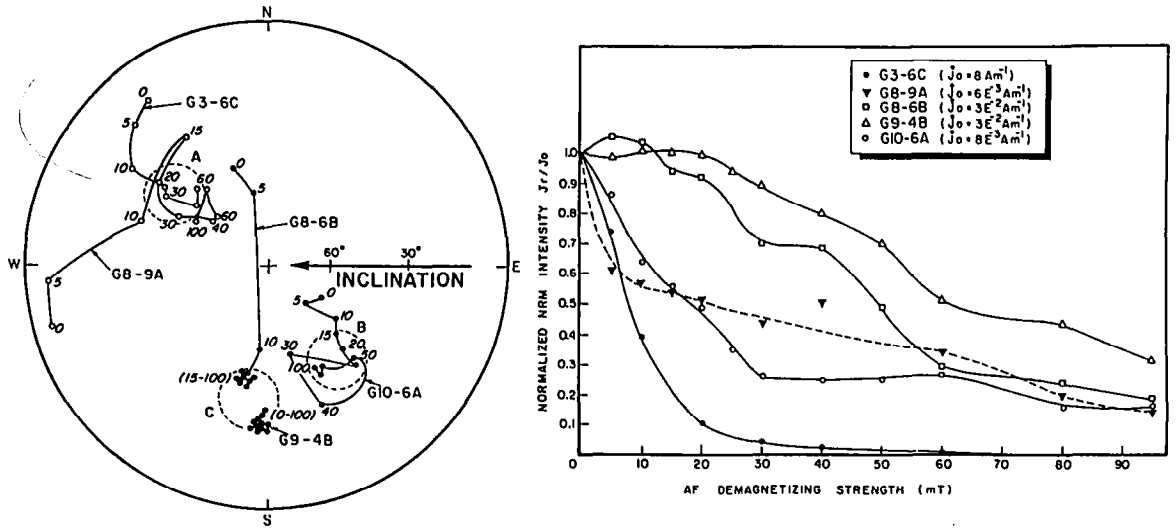


Fig. 3. Wulff net projections of directions of magnetization (left) of five specimens from five different samples (4 different sites) and depicting the three components (A, B and C) of magnetization after AF stepwise treatment (0–100 mT). The numbers next to the circles represent the demagnetization steps in mT. Solid (open) circles represent projection on the lower (upper) hemisphere. The respective normalized AF decay curves are shown on the right ride.

field (PEF). An ancient component of magnetization is preserved in these formations since an intermediate and negative (upward) northwest direction is frequently observed. The intensities range from 3×10^{-4} to 10^{-1} A m^{-1}.

5. Demagnetizations

Stepwise thermal and AF procedures were used up to 575°C and 100 mT respectively. Three components of magnetization were isolated with both AF and thermal demagnetization techniques; they are by order of importance: (A) NW, intermediate

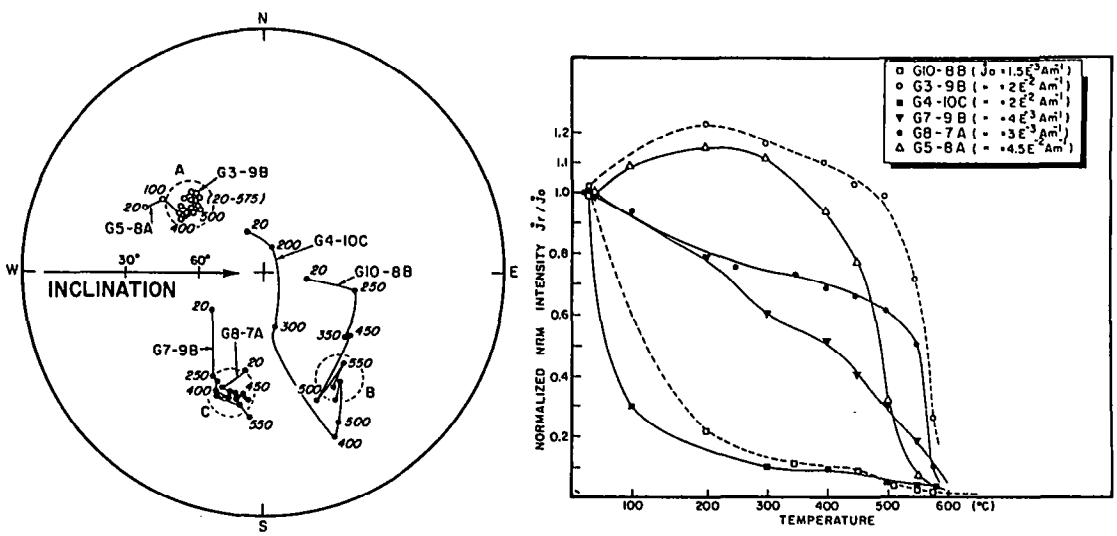


Fig. 4. Wulff net projections of directions of magnetization (left) of six specimens from six different samples (6 different sites) and depicting the three components (A, B and C) of magnetization after thermal treatment (20–575° or 600°C). The numbers next to the circles represent the demagnetization steps in °C. Solid (open) circles represent projections on the lower (upper) hemisphere. The respective normalized thermal decay curves are shown on the right side.

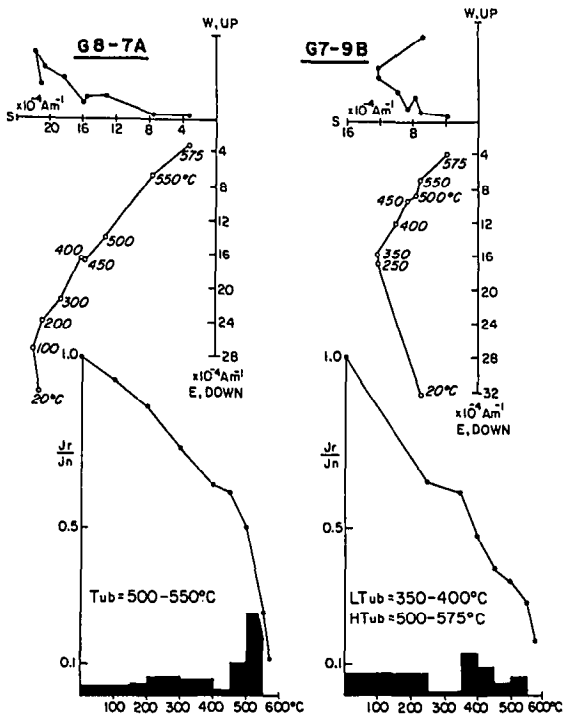


Fig. 5. Representative orthogonal thermal demagnetization diagrams [25] of successive endpoints of magnetization vector for two specimens from sedimentary rocks taken from two different sites (G7 and G8) as indicated. The numbers next to open circles represent the demagnetization steps in °C. Solid (open) circles represent projections on horizontal (vertical) plane. Units indicated for axes are in 10^{-4} A m^{-1} (10^{-7} emu cm^{-3}). This diagram illustrates the behavior of component C.

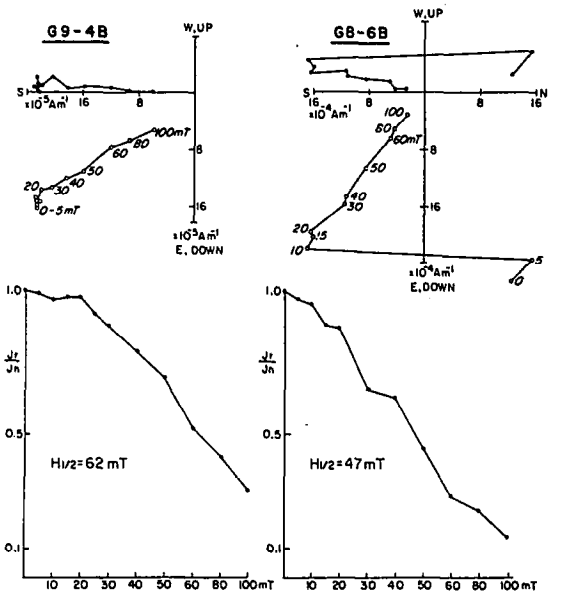


Fig. 6. Representative orthogonal AF demagnetization diagrams [25] of successive endpoints of magnetization vector for two specimens from sedimentary rocks taken from two different sites (G8 and G9) as indicated. The numbers next to open circles represent the demagnetization steps in mT. Solid (open) circles represent projections on horizontal (vertical) plane. Units indicated for axes are in 10^{-5} A m^{-1} (10^{-8} emu cm^{-3}). This diagram illustrates the behavior of component C.

upward inclination, (C) S, intermediate downward inclination, and (B) SE, intermediate downward inclination. The NW directions are characterized by a median destructive field ($H_{1/2}$) of 10–20 mT and unblocking temperature (T_{ub}) of 500–550°C. Specimens from sites G3 and G5 (Battery Point Formation) are examples of behavior of the A component upon AF (Fig. 3) or thermal (Fig. 4) demagnetizations. The SE direction of component B is characterized by the same $H_{1/2}$ and T_{ub} as component A (Figs. 3 and 4). Specimens G4-10C and G10-8B show an example of the 7 samples containing a very low-coercivity and low-temperature PEF component (Fig. 4). For the three directions of component C, $H_{1/2}$ values range from 40 to 80 mT and the T_{ub} values from 550 to 575°C. Fig. 5 shows two examples (specimens G8-7A and G7-9B) of the behaviour of the C component upon thermal demagnetization and Fig. 6 two other examples (specimens G9-4B and G8-6B) of its behaviour in the course of AF demagnetization. A low-temperature and low-coercivity component (e.g. specimen G8-6B in Fig. 3) which is located close to the PEF is occasionally overprinted on the C component. The T_{ub}, coercivity fields, observations with the aid of optical microscope (reflected light, ×850), electronic microscope and finally microprobe analyses indicate that the magnetic memory carrier is located in magnetite. Coarse-grained magnetite is often found in association with chromite grains; these opaque minerals are usually found in samples characterized by component B and occasionally component A. A very fine-grained magnetite which appears to be also of detrital origin is often detected in samples characterized by component C. The very fine grain of this oxide is suggestive of single or pseudo-single domain magnetite; this would explain the higher coercivity of component C. In summary, lower coercivities and somewhat lower T_{ub} are observed in component B especially in the samples overprinted by the PEF. Low to intermediate coercivi-

ties occur in component A and higher coercivities and sometimes T_{ub} are present in component C.

6. Mean direction of magnetization

Fig. 7 shows sample mean directions for the 10 sites. The directions were well determined from orthogonal plots [25]. Three groups are observed: one of normal polarity (A) in the NW quadrant, another of reverse polarity (B) in the SE quadrant and a third one (C) of reverse polarity in the S quadrant (Table 1). The reverse SE site-mean direction ($D = 327°$, $I = 30°$) is not significantly different ($\alpha_{95} = 12°$ and 8° respectively) from the NW site-mean direction ($D = 314°$, $I = -43°$). The site-mean direction of component C is $D = 182°$, $I = 41°$; this component is characterized by a larger dispersion but its occurrence is not negligible, particularly in view of its high coercivity and higher T_{ub} values compared with component A-B.

The application of statistical analysis (e.g. [26]) shows that the directions of components B and C are significantly different at a 95% confidence level. The normal polarity component A is isolated at four sites and only in a fraction of the

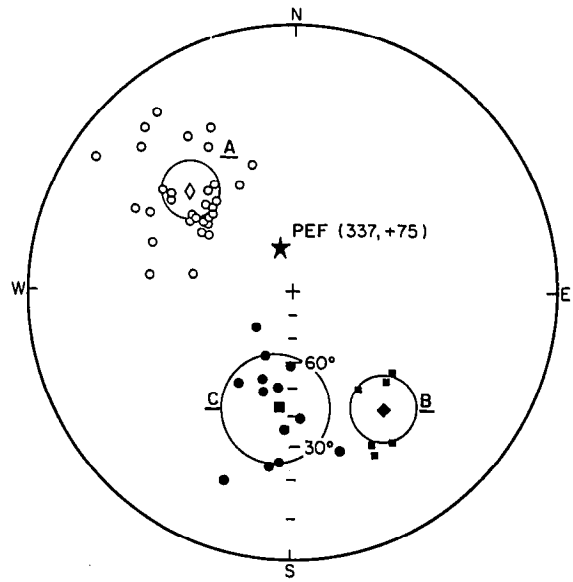


Fig. 7. Wulff net projection of sample-mean direction (open and closed circles) of magnetization obtained for the three components (A, B and C) in-situ. Diamonds and squares represent the in-situ site-mean direction for the same three components along with the errors circles (α_{95}). For comparison, the present Earth's field (PEF) is shown in the diagram.

TABLE 1

Gaspé sediments, site-mean remanence directions

Site	Component	N (n)	In-situ			Tilt-corrected			K	P
			D(°)	I(°)	α_{95}(°)	k	D(°)	I(°)		
G1	A	4 (7)	299	−44	18.5	25.6	306	−59		N
	B	2 (3)	144	37			155	48		R
G2	A	2 (7)	309	−52			348	−47		N
G3	A	7 (22)	311	−50	8.0	57.9	323	−51		N
G4	A	4 (7)	343	−34	27.2	10.5	348	−14		N
	B	1 (2)	153	32			159	14		R
G5	A	5 (16)	294	−49	13.7	31.9	351	−44		N
G6	A	3 (8)	317	−45	26.8	22.3	281	−51		N
G7	C	3 (7)	185	45	30.6	15.2	164	72		R
G8	A	2 (3)	309	−54			268	−74		N
	C	4 (8)	187	52	33.2	8.3	191	78		R
G9	C	6 (13)	181	33	16.5	23.4	174	80		R
G10	A	4 (7)	312	−26	17.0	30.2	295	−32		N
	B	3 (5)	142	45	17.7	49.6	120	52		R
Mean	A	8 (77)	314	−45	10.0	31.9	321	−50	9.4	N
	B	3 (10)	147	38	12.2	10.2	147	39	9.9	R
	A+B	11 (87)	318	−43	7.9	34.1	323	−47	10.0	N+R
	C	3 (28)	184	43	15.2	66.6	174	75	320	R

N = number of samples; n = number of specimens; D = declination; I = inclination; α_{95} = semi-axes of 95% confidence level; k = precision parameter; P = polarity: N (normal), R (reverse).

samples analysed (Table 1). At each of these sites either component B is detected in the remaining samples or component C, but never both. Nor is there a site with only components B and C. Reason(s) must be given to explain why both B and C components are never observed at a specific site together.

There may be different explanations for this situation. I think that the principal reason resides in the magnetic mineralogy of the sediments. A PEF component (when existing) is generally observed in component B only where coarse-grained magnetite (soft magnetization) is present. Component A is characterized by a slightly higher-coercivity spectra (a mixture of coarse-grained and very fine-grained (hard magnetization) magnetite). Consequently, component B which is the youngest component can be superimposed on component A. Component C is characterized by very fine-grained magnetite (hard magnetization). Its coercivity spectra is too high to be overprinted by component B but component A does overprint it partly.

7. Fold test

The results of the in-situ and tilt-corrected mean directions of magnetization are shown in Table 1. The degree of dispersion is substantially increased after tilt correction for components A and B; the fold test for component A and components A and B combined is significantly negative at the 95% confidence level [27]. As the deformation took place during the Acadian orogeny (Mid to Late Devonian), the age of magnetization is necessarily younger than this. The mean direction of magnetization ($D = 324°$, $I = -43°$) of the Carboniferous diabase dykes (300 Ma) being similar to that found in the sediments, it can be concluded that component A-B is secondary. For component C, the fold test is definitely positive but not quite significantly positive at the 95% confidence level ($k_2/k_1 = 5.9$ vs. $k_2/k_1 = 6.4$) using McElhinny's test [27]. However, using the McFadden and Jones' test [28] which is a little less restrictive, the fold test is positive for the same confidence level as above. Consequently, the tilt-corrected mean direction ($D = 174°$, $I = 75°$) is pre-folding, that is older than Late Devonian; it may be primary (i.e. related to deposition or diagenesis) or secondary. In any event, its time of acquisition must span the Lower/Middle Devonian interval.

8. Palaeomagnetic baked contact test

At each site investigated, the palaeomagnetic baked contact test (*pbct*) was applied. Fig. 8 shows examples of a negative and positive *pbct*. In the Indian Point and Battery Point Formations, the *pbct* is negative (e.g. site 3). In the York River Formation and one site in the Battery Point Formation (site G8 in Fig. 6), the *pbct* is most commonly positive. At site G3, the mean direction of magnetization in the sediments is the same as in the diabase dykes. This means that the sandstones were magnetically overprinted approximately at the time of intrusion and cooling of the dykes; the *pbct* is thus negative. At site G8, the direction of magnetization in the sediments is different from that of the dykes; however, at the contact diabase dykes–sandstone, the direction of magnetization shifts gradually from the SE to the S; this means that the magnetization of the dyke overprinted that of the sediment only near the contact. In other words, the *pbct* is positive. The *pbct* is always positive in the case of component C. The combination of a positive *pbct* with a positive fold test for component C demonstrates that this component is the oldest.

9. Comparison of the Devonian and Carboniferous Gaspé Peninsula poles with cratonic North Atlantic Devonian poles of the same age

The palaeopole obtained for the in-situ combined A and B components is 154°E, 7°N and the palaeolatitude 25°S (Fig. 9). This value agrees reasonably well with the pole (150°E, 18°N) obtained by Seguin et al. [29] for Late Devonian granites from the Ste. Cécile–St. Sébastien region, Quebec Appalachians; their palaeolatitudes are located between 20° and 25°S (Fig. 7). Consequently, the A-B component has a most probable Late Devonian age corresponding to the Acadian Orogenic event. The palaeopole obtained for the tilt-corrected C component (119°E, 20°S) is comparable to poles CB-1 (139°E, 16°S) and CB-2 (83°E, 32°S) obtained from Siluro-Devonian granitic rocks of Cape Breton Island [30], poles 73 and 80 shown in the compilation of Roy et al. [31]

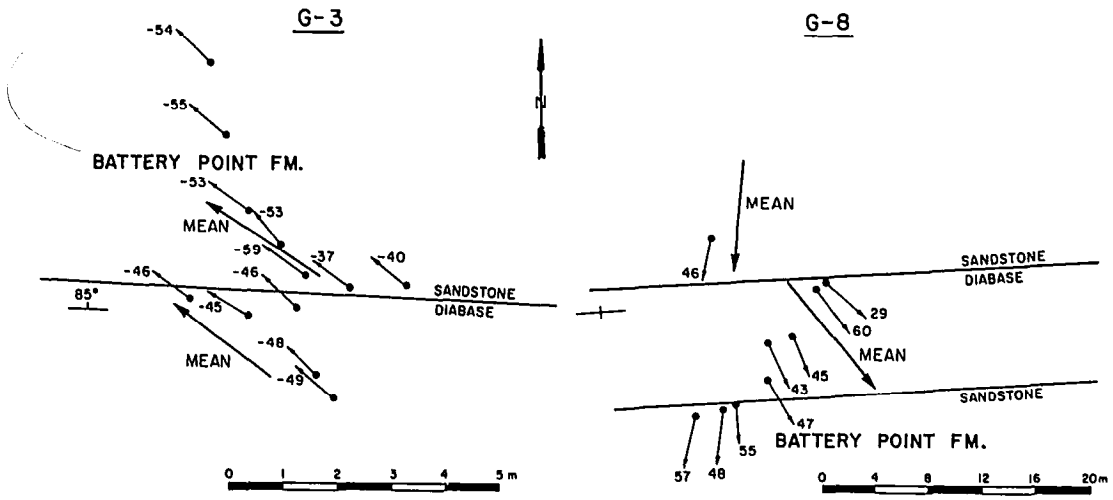


Fig. 8. Representative examples of the palaeomagnetic baked contact (*pbct*) test taken from two sites (G3 and G8). On site G3, the *pbct* is negative and on site G8 it is positive. Each dot represents a sampling location.The small arrows represent the direction of magnetization (declination) and the numbers the inclination with respect to the horizontal plane. The sample mean direction for the sedimentary rock and the diabase dyke is represented by the long arrows. On site G3, the *pbct* could be performed on only one side of the diabase dyke.

and pole DB-1 (109°E, 30°S) obtained from north central Newfoundland. A very Early Devonian age is thus consistent with the palaeopole position obtained in this study. A high southern palaeolatitude (60°S) is thus characteristic of the Siluro-Devonian and Lower Devonian time interval. These palaeomagnetic results imply a large motion of the APW path during the Siluro-Devonian to Lower-Middle Devonian timespan since little or no tectonic event (structural rotation) took place in this time interval. The palaeolatitude shifted from 60°S at Late Silurian time to 20°S at Late Devonian time, i.e. some 4000 km in 50 Ma or a minimum of 8 cm yr^{-1}, this is about twice as fast as the present rate of spreading for the East Pacific Rise.

Even though the proposed Devonian loop may be too large and/or partially misplaced, the occurrence of high palaeolatitudes during this period largely supports its existence. In the present study, it is indeed possible that the principle of minimum APWP (MAPWP) is not always respected. Given the density and the quality of the data set (Fig. 9), I am quite confident that for the Middle and Late Devonian period a MAPWP is defined. Given the paucity of data for the Early Devonian period, the same conclusion relative to a MAPWP cannot be arrived at. Still, it is noteworthy that at Siluro-De-

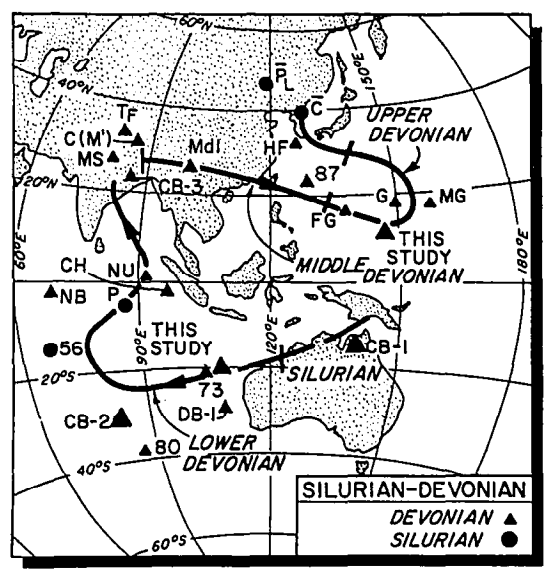


Fig. 9. Most reliable poles obtained from Devonian rock units from Northern Appalachians (after Seguin et al. [29] and Roy et al. [31]). The source of the poles shown on this diagram is as follows P_L and C (mean Lower Permian and Carboniferous poles) [31]; MS, FG, G, MG and HF [29]; 87 [32]); Nu, Md1 [33]; TF [3]; NB [10]; DB-1 [34]; 73 [31]; 56 [35]; P [36]; CB-1, CB-2 and CB-3 [30]; C (M') [37]; 80, JA [38]; CH [39]. The proposed APWP is represented by the black curve. The approximate Upper Devonian/Carboniferous boundaries are indicated.

vonian time, the poles for the Andean margin of Iapetus (e.g. 73, DB-1, 56, 80 and the Gaspé pole, Fig. 9) are practically on the same track as the Avalon poles (e.g. CB-1, CB-2, CH; Fig. 9). This suggests that at this time the Andean margin of Iapetus and the Avalon terrane(s) were already juxtaposed landmasses.

Acknowledgements

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